

MARION COUNTY AP, SC, USA

WMO: 720608

Lat: 34.181N

Lon: 79.335W

Elev: 92

StdP: 14.65

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00200

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.9	27.8	6.9	7.9	30.6	11.9	10.1	34.2	23.5	60.3	19.8	53.6	3.6	20	0.375

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	95.0	75.6	91.4	75.3	90.4	75.0	79.0	86.9	78.1	86.3	77.3	85.7	7.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.3	142.4	80.9	76.7	139.5	80.8	74.8	130.9	80.2	42.6	85.7	41.4	87.5	40.6	84.6	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	16.0	13.0	17.8	97.0	3.4	2.8	15.4	99.0	13.4	100.7	11.5	102.2	9.0	104.3
			15.7	80.3	3.6	1.1	13.0	81.1	10.9	81.7	8.9	82.3	6.2	83.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.7	45.2	48.8	54.1	63.3	71.4	78.1	80.7	79.4	74.9	64.5	53.1	49.9
	DBStd	14.47	10.09	9.29	9.33	7.32	6.08	4.51	3.44	3.16	4.81	7.36	8.03	8.68
	HDD50	581	216	127	65	4	0	0	0	0	0	2	54	113
	HDD65	2511	617	460	355	116	19	0	0	0	2	99	366	477
	CDD50	5572	65	92	192	403	663	843	952	910	747	450	146	109
	CDD65	2028	2	5	17	65	218	393	487	445	299	82	8	7
	CDH74	17964	8	36	164	609	1761	3694	4800	3923	2299	611	36	23
	CDH80	7187	0	1	19	128	585	1612	2218	1645	849	129	1	0
(14) Wind	WSAvg	5.2	5.6	6.2	6.5	6.6	5.5	4.7	4.8	4.1	4.2	4.7	4.5	4.9
(15) Precipitation	PrecAvg	48.0	3.7	3.2	3.6	2.9	3.6	4.8	5.4	6.2	5.0	3.5	3.1	3.1
	PrecMax	58.6	7.2	8.6	7.0	5.4	6.6	8.6	9.5	12.0	14.1	7.6	6.2	6.8
	PrecMin	33.4	1.2	1.3	0.8	0.4	1.1	1.1	2.3	3.4	0.9	0.1	0.2	1.0
	PrecStd	7.9	2.0	1.6	1.5	1.3	1.2	1.8	1.9	2.4	3.5	2.1	1.7	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.6	78.7	82.1	86.4	95.0	97.5	98.6	95.1	91.3	88.2	78.8	77.2
		MCWB	64.6	64.7	63.8	68.1	72.8	75.1	77.1	76.6	73.1	74.1	66.8	68.3
	2%	DB	70.1	73.1	79.0	82.5	89.6	93.5	95.2	91.3	90.2	82.4	73.1	72.6
		MCWB	62.2	62.8	63.0	65.8	71.2	75.0	76.2	76.2	73.3	70.0	63.7	65.6
	5%	DB	64.4	70.0	73.5	81.1	86.2	90.8	91.5	90.4	88.1	80.8	70.5	69.7
		MCWB	58.5	60.0	60.1	65.7	70.9	74.1	76.2	75.8	73.4	68.8	62.6	65.1
	10%	DB	62.6	64.3	70.3	77.3	82.2	88.4	90.4	88.4	85.9	77.3	68.0	64.4
		MCWB	56.8	57.2	59.4	63.7	69.7	73.6	76.1	75.3	73.3	66.6	61.9	60.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.4	67.0	67.8	73.1	75.7	78.8	80.9	80.3	78.3	75.4	70.1	70.3
		MCDB	71.0	73.6	74.3	79.1	86.1	89.6	89.7	87.4	83.4	83.4	73.4	73.9
	2%	WB	64.2	65.2	66.1	70.8	74.0	77.5	79.3	78.9	76.9	73.9	67.8	68.2
		MCDB	67.6	72.3	73.8	77.4	83.0	88.4	87.8	85.9	83.8	80.2	70.6	71.5
	5%	WB	61.7	62.6	64.1	69.0	72.8	76.4	78.2	78.0	75.6	72.1	65.3	65.3
		MCDB	64.8	67.6	70.4	75.3	81.2	86.5	87.0	85.2	83.4	77.0	70.1	68.2
	10%	WB	57.2	58.9	62.3	67.1	71.8	75.3	77.4	77.2	74.6	70.0	63.1	62.2
		MCDB	61.1	63.8	68.0	74.2	79.7	84.5	86.3	84.6	81.8	73.9	65.9	64.7
(35) Mean Daily Temperature Range	5% DB	MDBR	19.7	20.9	22.1	22.9	20.1	19.4	19.1	17.7	18.2	20.4	21.8	19.0
		MCDBR	24.7	24.9	25.9	25.9	23.5	21.8	20.9	20.0	21.3	22.3	23.7	21.4
		MCWBR	16.8	15.0	12.7	11.3	8.3	7.0	6.6	6.6	7.3	10.6	14.6	14.3
	5% WB	MCDBR	21.0	23.1	21.3	19.9	19.1	19.6	18.9	17.1	17.2	18.1	18.8	18.2
		MCWBR	16.8	16.1	12.2	10.6	7.8	7.4	7.1	6.7	6.8	10.7	14.0	14.2
(40) Clear-Sky Solar Irradiance	taub		0.319	0.322	0.362	0.401	0.465	0.489	0.538	0.509	0.430	0.392	0.343	0.322
	taud		2.506	2.494	2.407	2.285	2.162	2.138	2.022	2.092	2.307	2.380	2.475	2.530
	Ebn at Noon		283	295	289	281	263	256	243	247	264	265	271	275
	Edh at Noon		27	30	36	42	48	49	55	50	39	33	27	25
(44) All-Sky Solar Radiation	RadAvg		869	1074	1428	1799	1952	2029	1938	1747	1495	1245	957	764
	RadStd		49	111	117	121	151	135	120	86	123	146	77	62

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (63 neighbors)	+0.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+306	+220

Nomenclature: See separate page